

CSE 200 Annotated LaTeX Demo

Every construct from `prev_onlines/`, with commentary

Zarif Ishmam¹, Pierre-Simon Laplace², and Johannes Kepler³

¹Department of CSE, BUET, Dhaka, Bangladesh

²Académie des Sciences, Paris, France

³Imperial Mathematician, Prague

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1 Text formatting

This sentence has **bold text**, *italic text*, *emphasized text*, and underlined text. SMALL CAPS, typewriter, sans serif, roman and *slanted* also turn up.

outer italic inner stays italic back vs. *outer italic* inner flips upright *back*

Font sizes: tiny scriptsize footnotesize small normalsize large Large

LARGE huge Huge.

Quotes: “proper double quotes” and ‘single quotes’. Dashes: inter-word hyphen, a number range 1–9 (en dash), and an em dash—like this—for an aside.

Escapes: 100% & \$ # _ { } and \ ~ ^.

Inline verbatim: `\textbf{this is not bolded}` and 100% raw.

2 Colour

Colored text in red and a phrase in the custom blue defined in the preamble.

Everything inside these braces is red. Back to black.

A short highlighted note and a boxed and filled phrase.

40% blue on white, and teal darkened with 30% black.

In the time domain a system is described by a differential equation, whereas in the Laplace domain the same system is described by a rational algebraic expression.

The 19 predefined names are: red, green, blue, cyan, magenta, yellow, black, gray, white, darkgray, lightgray, brown, lime, olive, orange, pink, purple, teal and violet.

3 Lists

3.1 Nested and mixed lists

1. System Setup

- Install software packages
- Configure hardware
- ! Verify connections

2. Simulation Steps

- (a) Initialize parameters
- (b) Run simulation
 - Monitor CPU usage
 - Record intermediate results
- (c) Analyze outputs

3.2 Every label format the past papers use

- (I) Roman-numbered item
- (II) Another roman-numbered item
- (A) Capital-lettered item
- (i) Lowercase roman item

(ii) Another one

I Both of these items print the same label

I ...because I is literal, not `\Roman*`

Differential equations $\oplus$ initial conditions are included automatically;
 $\oplus$ derivatives become algebraic expressions.

Control and signal systems $\oplus$ stability is studied through poles;
 $\oplus$ common test inputs include:

- $\triangleright$ the unit-step input;
- $\triangleright$ the impulse input;
- $\triangleright$ sinusoidal inputs.

3.3 Symbol bullets

$\dagger$ `$\dagger$` – used by 23_C2

$\diamond$ `$\diamond$` – 23_C2, 23_A2, makeup

$\oplus$ `$\oplus$` – 23_B1, 23_A1

$\triangleright$ `$\rhd$` – 23_B1

$\nmid$ `$\nmid$` – 23_C1

$\ni$ `$\ni$` – 23_C1

$\ast$ `$\ast$`

$\star$ `$\star$`

$\circ$ `$\circ$`

$\bullet$ `$\bullet$`

$\rightarrow$ `$\rightarrow$`

$\#$ `\#` – a plain escaped character works too

3.4 Description lists and spacing control

Arc A continuous segment between two undercrossings.

Crossing A point where one projected strand passes over another.

Orientation A selected direction of travel around the knot.

- Widely spaced item, flush to the left margin
- Another one
 - Tight item
 - Another tight item

4 Spacing, breaks and alignment

Horizontal: a b, a b, a b, a b (thin), a b (forced). is also legal – 23_B1 uses it inside equations.

Left side

Right side

This paragraph is followed by `\vspace{0.8cm}`.

And this one comes after it.

A line break

comes before this text, so there is no indent.

This paragraph starts with `\noindent` so it is flush left.

This block is centred by the `center` environment.

This block is ragged right.

This block is ragged left.

5 Mathematics

5.1 Inline versus display

Inline maths: $x^2 + 1$, or with the newer delimiters $x^2 + 1$.

$$f(t) \xrightarrow{\mathcal{L}} F(s)$$

$$V_{\text{in}}(s) = \frac{V_0}{s}$$

$$L = I_{ij} \cdot \cos(\theta) + R^2 \tag{1}$$

$$c^2 = a^2 - b^2$$

5.2 Scripts, Greek letters and named operators

Superscripts and subscripts: x^2 , x_i , x_i^2 , x_{i_2} , e^{x^2} , T_M^2 , $\theta_i(t)$, $n^+(D)$, t^{-1} and x^{10} .

Lowercase Greek: $\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\vartheta\iota\kappa\lambda\mu\nu\xi\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$.

Uppercase Greek: $\Gamma\Delta\Theta\Lambda\Xi\P\Sigma\Upsilon\Phi\Psi\Omega$. (The ones that look like Latin letters – A, B, E – have no command; just type them.)

Named operators: $\sin\theta$, $\cos$, $\tan$, $\log$, $\ln$, $\exp$, $\min$, $\max$, $\gcd$, $\deg$, $\dim$, $\det(K)$, $\arg$.

Your own operator, with limits underneath: $\arg\max_x f(x)$.

5.3 Fractions, roots and sized delimiters

Fractions: $\frac{a}{b}$ (`frac`), $\frac{a}{b}$ (`dfrac`), $\frac{a}{b}$ (`tfrac`), $\frac{n!}{s^{n+1}}$, $\frac{1}{2}\sqrt{3}$.

Roots: $\sqrt{2x+1}$, $\sqrt[3]{x}$, $\sqrt[5]{2x+1}$.

Sized delimiters: $\left(\frac{1}{x^2+1}\right)$, $[\cdot]$, $\{\cdot\}$, $|\cdot|$, $\|v\|$, $\lceil\log_4 M\rceil$ (ceiling, 23_B2), $\lfloor(low+high)/2\rfloor$ (floor), and one-sided: $\frac{dy}{dx}\Big|_{x=1}$.

Fixed sizes when you do not want the automatic choice: $\left(\left(\left(\left(\right.\right.\right.\right.$

5.4 Sums, products, integrals and limits

$$\sum_{i=1}^n x_i \quad \prod_{p \text{ prime}} (1 - p^{-s})^{-1}$$

$$\int_0^\infty e^{-st} f(t) dt \quad \int_2^x \frac{dt}{\log t}$$

Other big operators: $\iint, \oint, \bigcup_i, \bigcap_i$.

Limits: $\lim_{x \rightarrow a} f(x)$ inline squashes, whereas $\lim_{x \rightarrow a} f(x)$ forces the limit underneath, and $\sum_{i=1}^n x_i$ forces the full display size.

5.5 Multi-line derivations: the four environments

$$\frac{x}{100} + 45 = 45 + \frac{4000 - x}{100} \tag{2}$$

$$2x = 4000, \quad x = 2000 \tag{3}$$

$$\begin{aligned} \mathcal{L}\{1\} &= \int_0^\infty e^{-st} dt \\ &= \left[-\frac{1}{s} e^{-st} \right]_0^\infty \\ &= \frac{1}{s}. \end{aligned} \tag{4}$$

$$\frac{T_M^2}{T_E^2} = \frac{a_M^3}{a_E^3}$$

$$T_M = T_E \left(\frac{a_M}{a_E} \right)^{3/2} \approx (1.524)^{3/2} \approx 1.88 \text{ years.}$$

$$\begin{aligned} RC[sV_C(s) - v_C(0)] + V_C(s) &= V_{in}(s), \\ \implies RCsV_C(s) + V_C(s) &= V_{in}(s), \\ \implies V_C(s)(RCs + 1) &= V_{in}(s). \end{aligned} \tag{5}$$

$$a = b + c \tag{6}$$

$$d = e \tag{7}$$

5.6 Piecewise functions

$$|y| = \begin{cases} y & \text{if } y \geq 0 \\ -y & \text{if } y < 0 \end{cases}$$

$$p(t) = \begin{cases} 0, & t < 0, \\ e^{-t}, & 0 \leq t < 3, \\ \ln(t-2), & t \geq 3 \end{cases}$$

$$\sigma(K) = \begin{cases} 0, & \text{if } K \text{ is the unknot,} \\ -2, & \text{if } K \text{ is the right-handed trefoil,} \\ 2, & \text{if } K \text{ is the left-handed trefoil.} \end{cases} \tag{8}$$

5.7 Matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \left\{ \begin{matrix} a & b \\ c & d \end{matrix} \right\} \quad \left| \begin{matrix} a & b \\ c & d \end{matrix} \right| \quad \left\| \begin{matrix} a & b \\ c & d \end{matrix} \right\| \quad \begin{matrix} a & b \\ c & d \end{matrix}$$

$$R = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2. \quad (9)$$

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

$$\left(\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right)$$

5.8 Math alphabets, text in maths, and decorations

Alphabets: $\mathbb{R} \mathbb{Z} \mathbb{Q} \mathbb{N} \mathbb{C}$ (blackboard bold, needs amsfonts); $\mathcal{L} \mathcal{A} \mathcal{S}$ (calligraphic – $\mathcal{L}$ is the Laplace transform symbol); $\mathbf{v} \mathbf{A}$ (bold upright, for vectors and matrices); $\mathrm{d}x$ and V_{in} (upright roman inside maths – subscripts that are WORDS should always be upright); x , $\mathbf{X}$, **code**; and ***θ*** for bold Greek.

Words inside maths: $\mathrm{Enc}(x_i)$ and $\{p : p \text{ is prime}\}$.

Accents and decorations: $\bar{x}$, $\hat{x}$, $\tilde{x}$, $\vec{v}$, $\dot{x}$, $\ddot{x}$, $\overline{AB}$, $\overrightarrow{ABC}$, $\overrightarrow{AB}$, $\underbrace{a+b+c}_{\text{three terms}}$, and the labelled arrow

$\xrightarrow{\mathcal{L}}$ used in 23_B1.

5.9 Symbol reference

Relations: $\leq \geq \neq \approx \equiv \sim \simeq \propto \ll \gg$

Arrows: $\rightarrow \twoheadrightarrow \Rightarrow \Longrightarrow \iff \leftarrow \longrightarrow \mapsto \Uparrow \Downarrow$

Sets and logic: $\in \notin \subseteq \supseteq \cup \cap \setminus \emptyset \forall \exists \neg \wedge \vee$

Operators: $\pm \mp \times \div \cdot \ast \oplus \otimes \circ \bullet$

Miscellaneous: $\infty \partial \nabla \angle 90^\circ \dagger \ddagger \odot M_\odot$ (solar mass, 23_C2).

Spacing inside maths, thinnest to widest: ab (negative), ab (none), $a\,b$ (thin), $a\,b$ (medium), $a\,b$ (thick), $a\ b$, $a\ b$, $a\ b$.

6 Theorems, propositions and proofs

Proposition 1 (Weak-field circular orbit). *For a circular orbit of radius r ,*

$$\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}.$$

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$. Divide by c^2 and use $r_s = 2GM/c^2$. $\square$

Theorem 6.1 (Prime-distribution error bound). *The Riemann Hypothesis is equivalent to $\psi(x) = x + O(\sqrt{x} \log^2 x)$.*

Lemma 6.2. *A helper statement that will be used to prove the theorem above.*

Definition 1. A *knot* is a closed curve in three-dimensional space with no self-intersections.

Cross-references work exactly as for figures: see Proposition 1, Theorem 6.1 and Lemma 6.2.

7 Tables

7.1 A plain boxed table

Table 1: Comparison of image-synthesis techniques.

Method	Accuracy	Time Complexity
Rasterization	Medium	$O(n)$
Ray Tracing	High	$O(n^2)$
Hybrid	Very High	$O(n^3)$

The results in Table 1 show the accuracy/cost trade-off.

7.2 The booktabs look

Table 2: Colour used to communicate status in a table.

Item	Result	Comment
Figure reference	Correct	Label placed after caption
Image size	Needs work	Width too large
Citation	Correct	Source appears in bibliography

7.3 multirow + multicolumn + cline: the exam table

Table 3: Selected Laplace-transform equations (the 23_B1 table).

Family	Transform Equation		Condition
	Time function	Laplace transform	
Algebraic	1	$\frac{1}{s}$	$s > 0$
	t	$\frac{1}{s^2}$	$s > 0$
	t^n	$\frac{n!}{s^{n+1}}$	$n = 0, 1, 2, \dots$
Exponential	e^{at}	$\frac{1}{s-a}$	$s > a$
	e^{-at}	$\frac{1}{s+a}$	$s > -a$
Oscillatory	$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$	$\omega > 0$
	$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$	$\omega > 0$
Unit-step function $u(t-a)$		$\frac{e^{-as}}{s}$	$a > 0$

7.4 Two multicolumn groups side by side

Table 4: Orbital and relativistic parameters of the outer planets.

Planet	Orbital parameters		Relativistic notation	
	a (AU)	e	$\varepsilon = \frac{1}{ac^2}GM_\odot$	$\beta = \sqrt{\varepsilon}$
Jupiter	5.204	4.89×10^{-2}	1.90×10^{-9}	4.36×10^{-5}
Saturn	9.583	5.65×10^{-2}	1.03×10^{-9}	3.21×10^{-5}
Uranus	19.191	4.72×10^{-2}	5.14×10^{-10}	2.27×10^{-5}
Neptune	30.070	8.6×10^{-3}	3.28×10^{-10}	1.81×10^{-5}

7.5 Colouring tables

Table 5: Row colour, cell colour and a wrapping paragraph column.

Term	Description
$f(x)$	A long description that needs to wrap onto several lines, which is why this column is declared <code>p{6cm}</code> rather than <code>c</code> .
Status	Correct

Table 6: Automatic zebra striping with `rowcolors`.

Item	Result	Comment
Alpha	Correct	Short note.
Beta	Needs work	A longer comment that wraps inside the paragraph column.
Gamma	Correct	Another note.
Delta	Correct	And one more.

8 Figures

8.1 A single centred figure

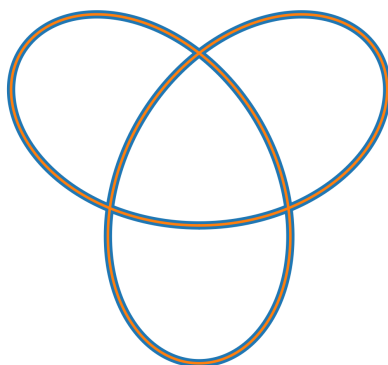


Figure 1: A planar drawing of a trefoil knot.

The simplest nontrivial knot is shown in Figure 1.

8.2 Width, scale and rotation in one figure

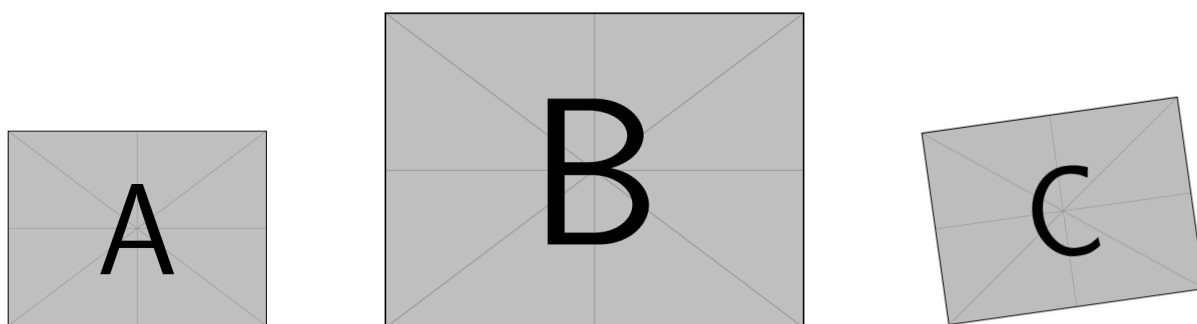


Figure 2: The same figure demonstrating **width**, **scale** and **angle**.

8.3 Float placement options

8.4 Subfigures in a row

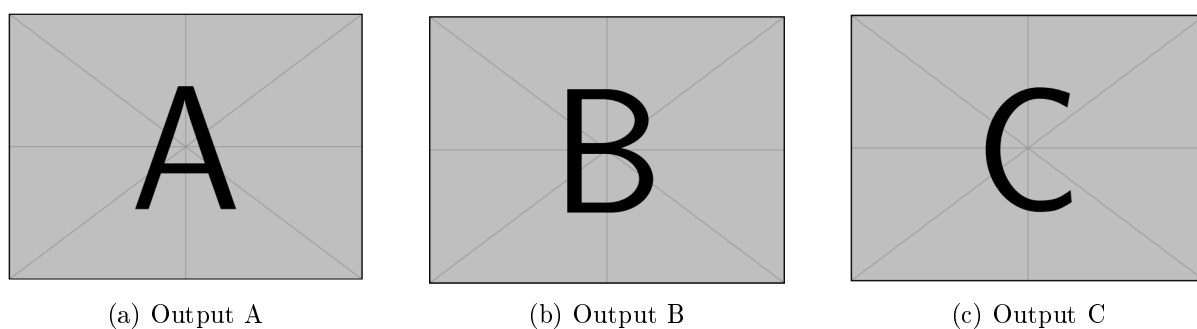
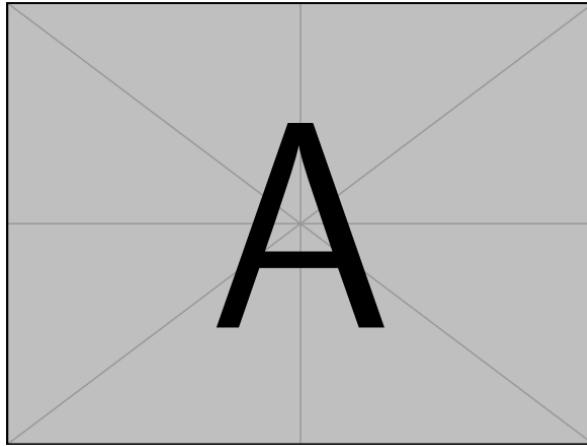


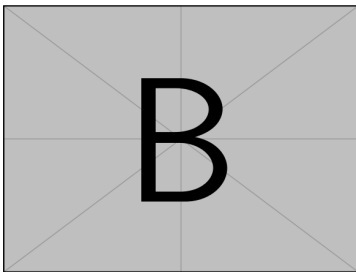
Figure 3: Comparison of three rendering outputs (the 22_A2 figure).

Figure 3 has three panels; compare Figure 3a with Figure 3c.

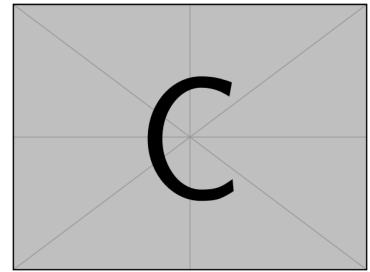
8.5 Asymmetric layout: one dominant figure above two smaller ones



(a) Main diagram



(b) Detail B



(c) Detail C

Figure 4: Asymmetric layout with one dominant figure (the 22_C2 figure).

8.6 Nine subfigures in labelled row groups

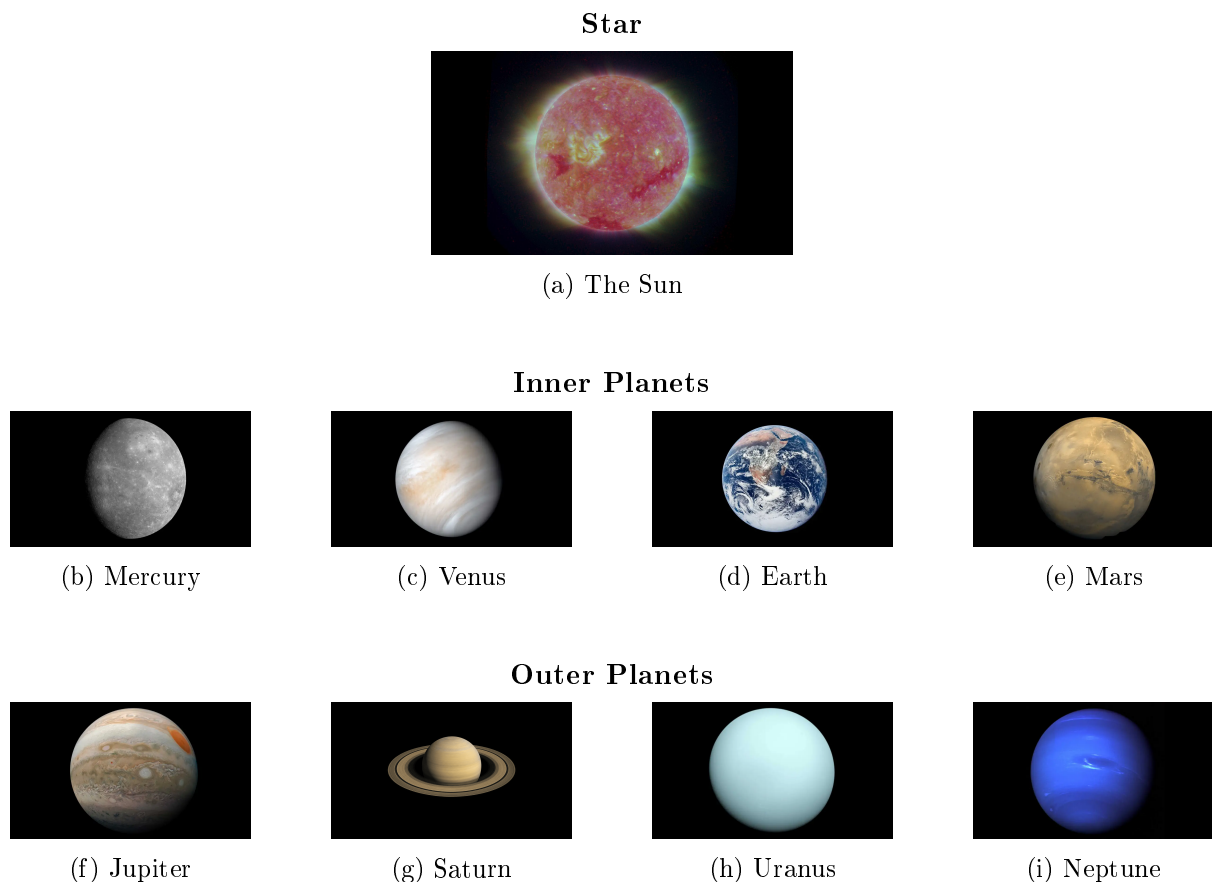


Figure 5: The Sun and the eight planets, arranged into the inner and outer planetary groups. The displayed sizes and distances are not to scale.

Compare Mars in Figure 5e with Jupiter in Figure 5f.

8.7 Wrapped figures

Figure 6 shows a simple resistor–capacitor circuit connected to an input voltage source. The resistor limits the current, while the capacitor stores electrical energy. For a resistor R and a capacitor C , Kirchhoff’s voltage law gives the governing differential equation $RC \dot{v}_C(t) + v_C(t) = v_{in}(t)$. Taking the Laplace transform with $v_C(0) = 0$ turns that differential equation into an algebraic one, and the transfer function follows immediately as $H(s) = 1/(RCs + 1)$. The quantity $\tau = RC$ is called the time constant of the circuit; a larger τ makes the capacitor charge more slowly. Notice how this paragraph has to keep going for several lines: that is deliberate, because a wrapped figure needs enough text beside it to fill the space it reserved.

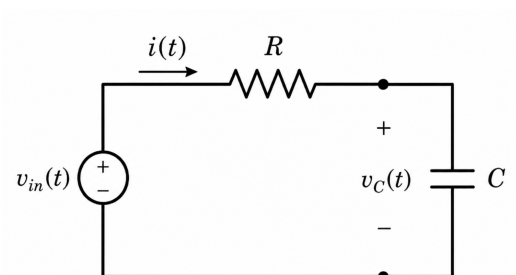


Figure 6: A simple series RC circuit.

8.8 Spanning both columns

9 Code listings

Listing 1: Computing the writhe from signed crossings (23_A1).

```
1 def writhe(crossing_signs):
2     """Return the writhe of an oriented knot diagram."""
3     if any(sign not in (-1, 1) for sign in crossing_signs):
4         raise ValueError("Each sign must be +1 or -1")
5     return sum(crossing_signs)
6
7
8 if __name__ == "__main__":
9     signs = [1, 1, -1, 1]
10    print(writhe(signs)) # Output: 2
```

The definition in Equation (1) and the code in Listing 1 perform the same summation.

```
@book{sample2024,
      author    = {Author Name},
      title     = {Book Title},
      publisher = {Publisher},
      year      = {2024}
}
```

10 Algorithms and pseudocode

Algorithm 1 Binary Search

Require: Sorted array A of length n , target value x

Ensure: Index of x , or -1 if not found

```
1:  $low \leftarrow 1, high \leftarrow n$ 
2: while  $low \leq high$  do
3:      $mid \leftarrow \lfloor (low + high)/2 \rfloor$ 
4:     if  $A[mid] = x$  then
5:         return  $mid$ 
6:     else if  $A[mid] < x$  then
7:          $low \leftarrow mid + 1$ 
8:     else
9:          $high \leftarrow mid - 1$ 
10:    end if
11: end while
12: return  $-1$ 
```

Algorithm 2 Every other control structure in one place

Require: Phases θ_i , frequencies ω_i , coupling K , step h , steps M **Ensure:** Updated phases

```
1: for  $m \leftarrow 1$  to  $M$  do
2:   for all fireflies  $i$  do
3:      $q_i \leftarrow \sum_{j=1}^N \sin(\theta_j - \theta_i)$ ,  $v_i \leftarrow \omega_i + \frac{K}{N}q_i$   $\triangleright$  one pass over all neighbours
4:   end for
5:    $\theta_i \leftarrow (\theta_i + hv_i) \bmod 2\pi$ 
6: end for
7: repeat
8:   adjust the flows
9: until  $T_{p^-}(f) - T_{p^+}(f) \leq \varepsilon$ 
10: function ADD( $a, b$ )
11:   return  $a + b$ 
12: end function
13: return  $\theta_1, \dots, \theta_N$ 
```

Algorithm 1 runs in

$$T(n) = \Theta(\log n)$$

time, while Algorithm 2 costs $T(M, N) = \Theta(MN^2)$.

11 Cross-references, footnotes and hyperlinks

11.1 Cross-references

Section 11.1 is this section, on page 14. Equation 1 is the same as Equation (1), but `\eqref` adds the parentheses for you. See also Table 3, Figure 5, Listing 1 and Algorithm 1.

11.2 Footnotes

The transform changes a function of time into a function of the new variable s .¹

First mention.² Second mention, method (a).² Third mention, method (b).²

A mark placed here³ with its text supplied afterwards.

11.3 Hyperlinks

A concise overview is available from the [Laplace Transform entry on MathWorld](#), and the raw address is <https://www.overleaf.com>. You can also link an email: [write to me](#).

12 Citations and the bibliography

The method replaces calculus in the time domain with algebra in the transform domain [1]. Vaswani et al. [2] introduced the Transformer architecture. Synchronisation in firefly swarms has been measured directly [3]. Two sources at once: [1, 2]. Vaswani et al. published it in 2017.

¹The variable s is commonly interpreted as a complex-frequency variable.

²A note that will be referred to twice.

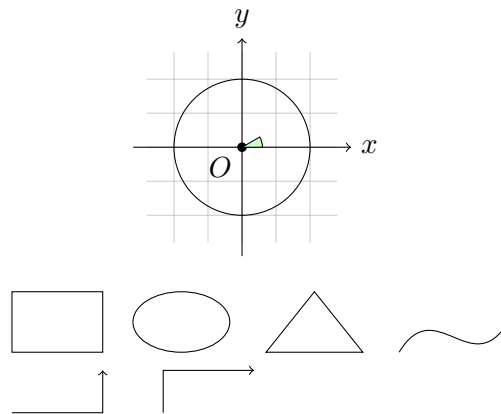
³This body was supplied by `\footnotetext`.

References

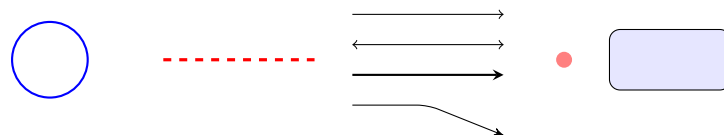
- [1] E. Kreyszig. *Advanced Engineering Mathematics*. John Wiley & Sons, 10th edition, 2011.
- [2] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.
- [3] Raphaël Sarfati, Julie C. Hayes, and Orit Peleg. Self-organization in natural swarms of *Photinus carolinus* synchronous fireflies. *Science Advances*, 7(28):eabg9259, 2021. doi: 10.1126/sciadv.abg9259.
- [4] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd edition, 1994.

13 TikZ

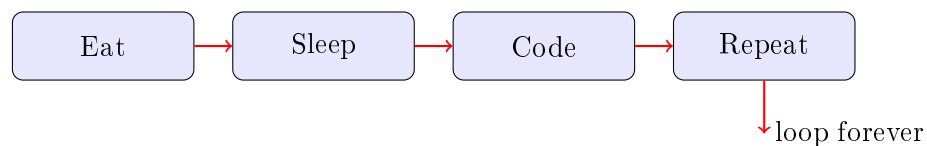
13.1 Basic paths



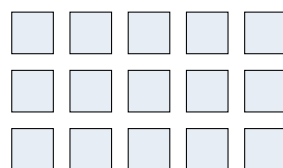
13.2 Options: colour, thickness, arrows, fill

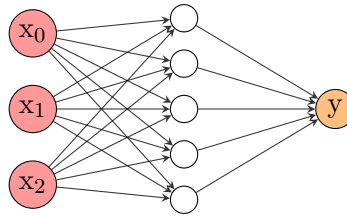


13.3 Nodes and relative positioning

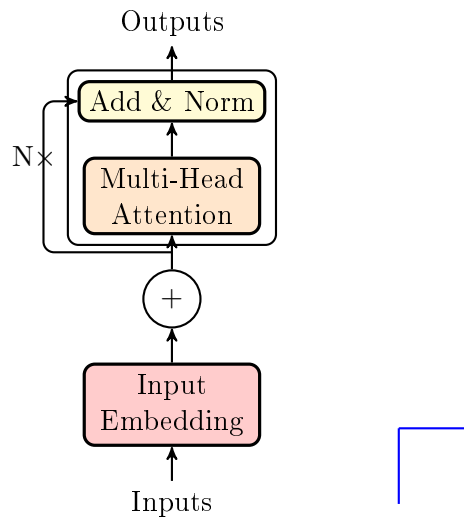


13.4 \foreach loops





13.5 Styles, scopes and the calc library



Inline TikZ: 

14 Two-column layout

14.1 Text in two columns

This paragraph is set in two-column mode. Everything you have seen so far – lists, maths, tables, figures – behaves identically here; the only difference is that `\linewidth` is now the width of one column rather than the width of the page, which is exactly why sizing images with `\linewidth` rather than `\textwidth` is the safer habit.

A normal **figure** or **table** float stays inside its column. To span both columns you need the starred forms, and they can only be placed at the top of a page.

Table 7: A table inside one column.

n	n^2
1	1
2	4
3	9

An equation also narrows to the column:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}. \quad (10)$$

Equation (10) converges for $\operatorname{Re}(s) > 1$, and Table 7 sits happily inside a single column.

Back to one column

That is every construct from the nine past papers. If you can reproduce this file from scratch, you can reproduce any of them.